# Title:

# Quantitative Estimation of Salivary Amylase Activity Using DNS Method

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**Course:** BSc Biotechnology

**Focus Area:** Data Analysis in Enzyme Assays

## Objective:

To estimate the activity of salivary amylase by measuring maltose production using the DNS method and analyzing results through a standard calibration curve.

## Background

Salivary amylase is a digestive enzyme responsible for the hydrolysis of starch into smaller sugars such as maltose. The activity of this enzyme can be quantified by measuring the amount of reducing sugar released during the reaction.

The DNS (3,5-dinitrosalicylic acid) method is commonly used for detecting reducing sugars because it produces a coloured complex when reacting with aldehyde groups of sugars. The intensity of this colour is measured spectrophotometrically at **540 nm**, allowing quantitative estimation of maltose concentration.

## Methodology

**Step 1:** Preparation of maltose standard solutions.

**Step 2:** Reaction of maltose with DNS reagent to produce a coloured complex.

**Step 3:** Measurement of absorbance at **540 nm** using a spectrophotometer.

**Step 4:** Construction of a standard calibration curve of maltose concentration vs absorbance.

**Step 5:** Measurement of absorbance values from salivary amylase samples.

**Step 6:** Calculation of maltose produced using the regression equation obtained from the calibration curve.

## Data Analysis Workflow

1. Experimental absorbance values were recorded using a spectrophotometer at **540 nm** after reaction with DNS reagent.
2. A **maltose calibration curve** was constructed using known concentrations of maltose.
3. Linear regression analysis was performed to obtain the calibration equation.
4. The regression model was used to calculate the **maltose concentration produced by salivary amylase samples**.
5. Data visualization was performed using **scatter plots and bar graphs in Microsoft Excel**.

## Standard Curve Data

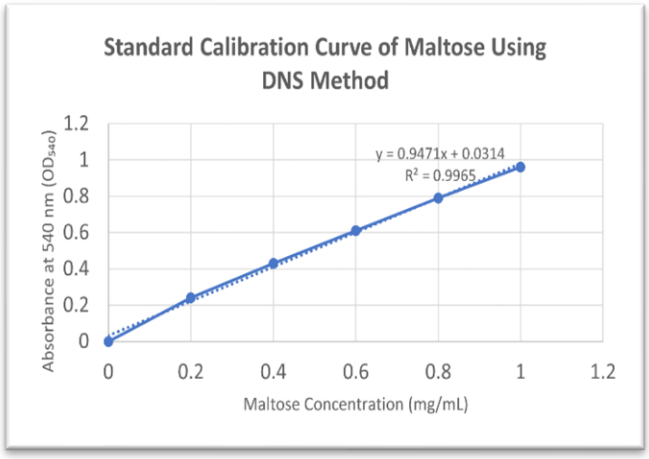
| Maltose Concentration (mg/mL) | Absorbance (OD540) |
|---|---|
| 0 | 0 |
| 0.2 | 0.24 |
| 0.4 | 0.43 |
| 0.6 | 0.61 |
| 0.8 | 0.79 |
| 1.0 | 0.96 |

**Regression Equation:**

$y = 0.9471x + 0.0314$

**Correlation Coefficient:**

$R^2 = 0.9965$



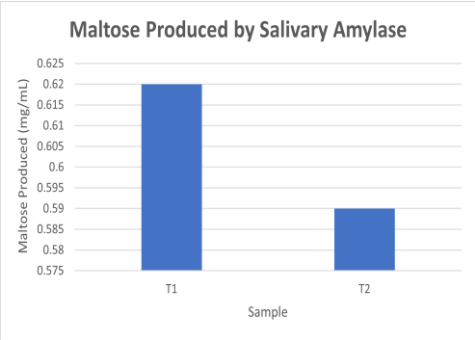

## Salivary Amylase Activity Data

| Sample | Absorbance (OD540) |
|---|---|
| Control | 0 |
| T1 | 0.62 |
| T2 | 0.59 |

## Calculated Maltose Production

Using the regression equation from the calibration curve, the maltose concentration produced by salivary amylase was estimated.

| Sample | Maltose Produced (mg/mL) |
|---|---|
| T1 | 0.62 |
| T2 | 0.59 |



## Data Quality Assessment

- The calibration curve showed a strong linear relationship between maltose concentration and absorbance.
- The regression coefficient **$R^2$ = 0.9965** indicates high reliability of the analytical model.
- Replicate enzyme samples (T1 and T2) produced similar maltose values, suggesting **good experimental reproducibility**.
- No absorbance was detected in the control sample, confirming **valid assay conditions**.

## Results and Interpretation

The calibration curve showed a strong linear relationship between maltose concentration and absorbance, with an $R^2$ value of **0.9965**, indicating high reliability of the analytical method.

The salivary amylase samples produced measurable amounts of maltose, confirming enzymatic hydrolysis of starch. The slight variation between T1 and T2 suggests consistent enzyme activity and reproducibility of the experiment.

## Applications

The estimation of reducing sugars using the DNS method has several important applications in biotechnology and biochemical research. This method is widely used to study the activity of digestive enzymes such as salivary amylase by measuring the amount of maltose produced during starch hydrolysis.

In food biotechnology, the DNS assay is applied to monitor carbohydrate breakdown during fermentation and food processing. It is also useful in enzyme kinetics studies to determine reaction rates and enzyme efficiency under different conditions.

Additionally, similar analytical approaches are used in biochemical laboratories for quantitative analysis of sugars and metabolic products. The use of calibration curves and

spectrophotometric measurements helps ensure accurate and reproducible data, which is essential in scientific research and laboratory data management.

## Conclusion

The experiment successfully demonstrated the determination of salivary amylase activity using the DNS method. A standard calibration curve of maltose concentration versus absorbance was constructed, which showed a strong linear relationship with a high correlation coefficient ($R^2 = 0.9965$).

The absorbance values obtained from the salivary amylase samples confirmed the enzymatic hydrolysis of starch into maltose. The use of a calibration curve enabled accurate estimation and interpretation of the experimental data.

Overall, the DNS method proved to be a reliable and effective technique for estimating reducing sugars and analyzing enzyme activity. This experiment also highlights the importance of proper data collection, graphical representation, and quantitative analysis in biochemical research.

## Skills Demonstrated

- Experimental data collection
- Calibration curve analysis
- Quantitative biochemical analysis
- Data visualization using graphs
- Scientific documentation